

Deep Convolutional Networks for ImageNet Classification

Slidev with Slide Generation System

Table Of Content

1. ImageNet Dataset and Challenge Context
 - 1.1. Large-scale ImageNet Collection
 - 1.2. ILSVRC Competition Metrics
2. Deep Convolutional Network Architecture
 - 2.1. ReLU Nonlinearity and GPU Parallelization
 - 2.2. Local Response Normalization and Pooling

Table Of Content

3. Strategies to Prevent Overfitting

3.1. Data Augmentation Techniques

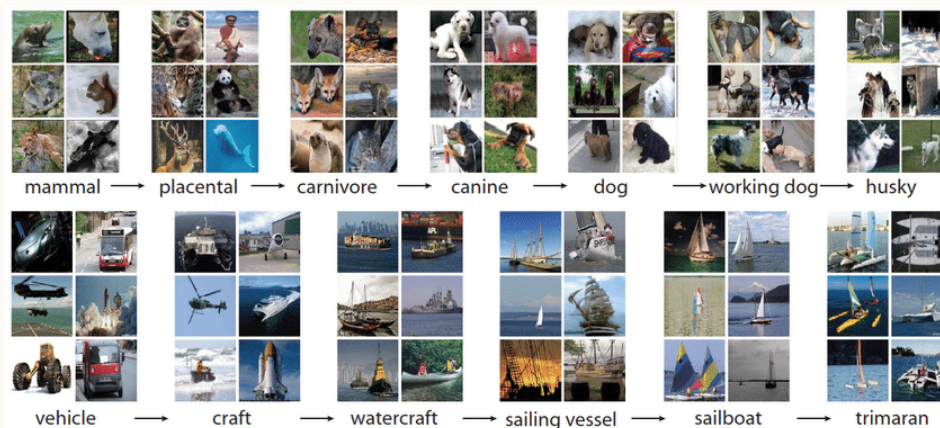
3.2. Dropout Regularization Method

4. Experimental Results and Analysis

4.1. Performance Comparison with State-of-the-Art

4.2. Qualitative Evaluation of Learned Features

1. ImageNet Dataset and Challenge Context



ImageNet dataset visualization showing diverse object categories and labeled high-resolution image examples

- ImageNet contains over 15 million labeled high-resolution images across 22,000 categories.
- ILSVRC uses a subset of 1.2 million training images across 1000 distinct object classes.
- Images are down-sampled to a fixed 256×256 resolution with mean subtraction applied.
- The central 256×256 patch is cropped from the rescaled rectangular input images.
- Performance is measured using top-1 and top-5 error rates on the test set.

1.1 Large-scale ImageNet Collection



ImageNet dataset visualization showing millions of labeled high-resolution images across diverse object categories

- ImageNet contains over 15 million labeled high-resolution images across roughly 22,000 categories.
- Images were collected from the web and labeled by humans using Amazon's Mechanical Turk.
- The ILSVRC subset used for training includes roughly 1.2 million images in 1,000 categories.
- This massive scale enables learning thousands of objects, overcoming limitations of smaller datasets.

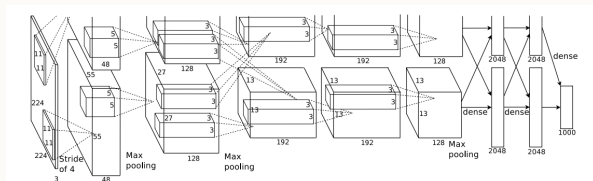
1.2 ILSVRC Competition Metrics

Model	Top-1	Top-5
<i>Sparse coding [2]</i>	47.1%	28.2%
<i>SIFT + FVs [24]</i>	45.7%	25.7%
CNN	37.5%	17.0%

Table 1: Comparison of results on ILSVRC-2010 test set. In italics are best results achieved by others.

- Our CNN achieves a top-1 error of 37.5% and top-5 error of 17.0% on the ILSVRC-2010 test set.
- This outperforms the previous best SIFT + FVs approach by 8.7 percentage points in top-5 error.
- We also surpass sparse coding models, which recorded 47.1% top-1 and 28.2% top-5 errors.
- These results represent a significant improvement over all previously reported state-of-the-art methods.
- The deep convolutional architecture effectively handles the 1.2 million high-resolution images in the dataset.

2. Deep Convolutional Network Architecture



Two-GPU CNN architecture with specific layer distribution and neuron counts.



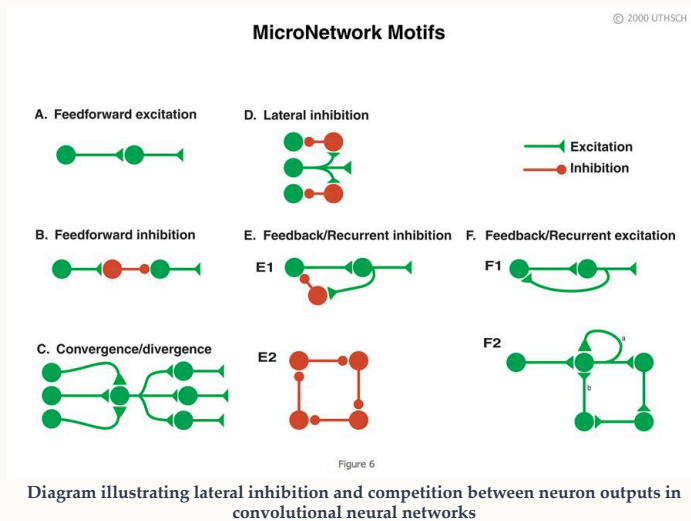
96 11x11x3 kernels from the first layer, split across two GPUs.

- The network contains eight learned layers: five convolutional and three fully-connected.
- A 1000-way softmax at the output produces a distribution over class labels.
- We spread the network across two GPUs to handle memory constraints of 3GB devices.
- Training on two GPUs reduces top-1 and top-5 error rates compared to single-GPU nets.
- The network size is limited mainly by available GPU memory and training time tolerance.

2.1 ReLU Nonlinearity and GPU Parallelization

ReLU ($f(x) = \max(0, x)$) train several times faster than saturating neurons like $\tanh$ or sigmoid. Figure 1 shows ReLU networks reach 25% training error six times faster than equivalent $\tanh$ networks. Saturating nonlinearities suffer from slow gradient descent, whereas ReLUs avoid saturation and accelerate learning. This speedup was essential for experimenting with the large-scale networks required for ImageNet.

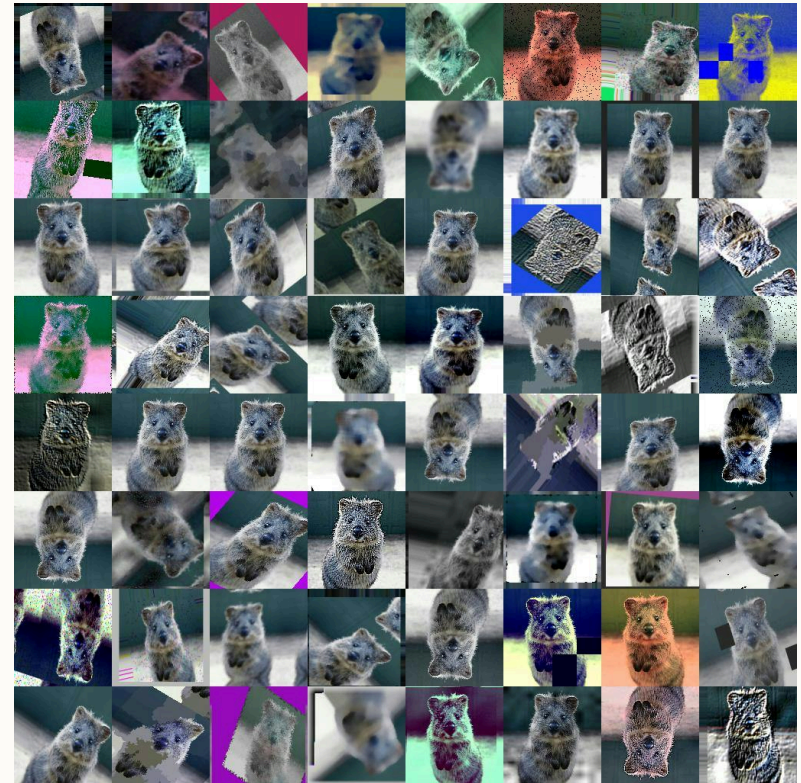
2.2 Local Response Normalization and Pooling



- LRN implements lateral inhibition, creating competition among neuron outputs from different kernels.
- Hyper-parameters $k=2$, $n=5$, $\alpha=10^{-4}$, and $\beta=0.75$ were tuned on a validation set.
- Adding LRN reduced top-1 and top-5 error rates by 1.4% and 1.2% respectively.
- Overlapping pooling ($s=2$, $z=3$) reduces error rates by 0.4% and 0.3% compared to non-overlapping schemes.
- Models with overlapping pooling find it slightly more difficult to overfit during training.

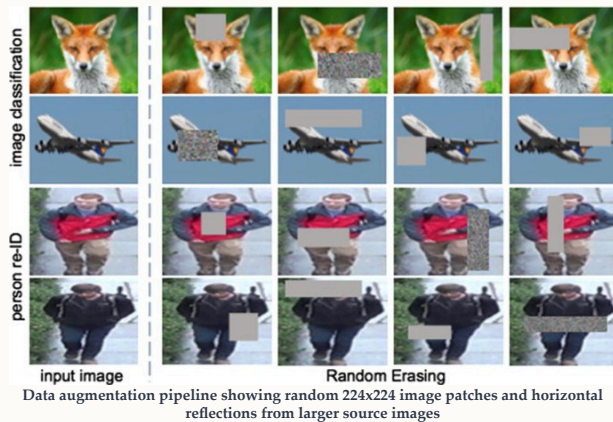
3. Strategies to Prevent Overfitting

- Overfitting is a significant problem for our 60 million parameter network despite 1.2 million training images.
- We employ data augmentation to artificially enlarge the dataset using label-preserving transformations like translations and reflections.
- We alter RGB channel intensities using PCA to capture invariance to illumination changes in natural images.
- We utilize a regularization method called 'dropout' to reduce complex co-adaptations of neurons in fully-connected layers.
- Dropout forces neurons to learn robust features useful with many different random subsets of other neurons.
- Without these techniques, our network suffers from substantial overfitting that would force us to use smaller networks.



Data augmentation techniques for image classification including translations reflections and color intensity PCA

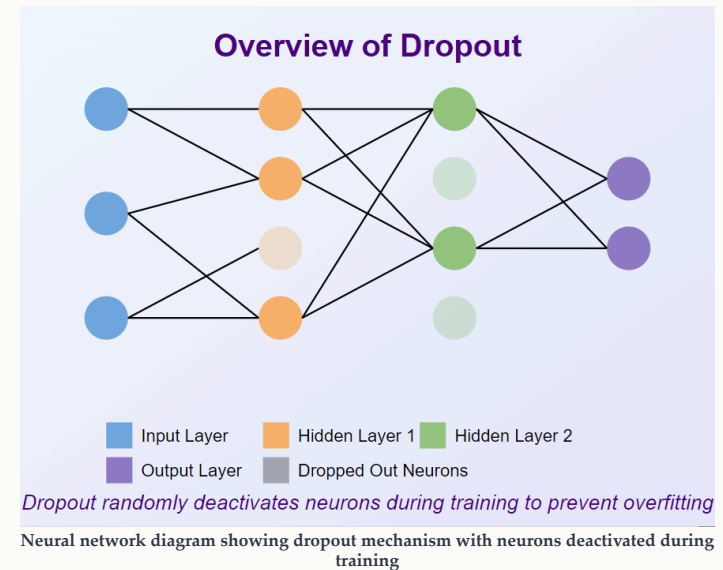
3.1 Data Augmentation Techniques



- Generate random 224×224 patches and their horizontal reflections from 256×256 images.
- This translation and reflection scheme increases the training set size by a factor of 2048.
- Perform PCA on RGB pixel values to add multiples of principal components with Gaussian noise.
- This color jittering captures object identity invariance to changes in illumination intensity and color.
- Transformed images are generated on the CPU while the GPU trains, making augmentation computationally free.
- At test time, average predictions from ten patches (five crops and their reflections) for final output.

3.2 Dropout Regularization Method

- Dropout sets each hidden neuron's output to zero with probability 0.5 during training.
- Dropped neurons do not contribute to the forward pass or participate in back-propagation.
- Every training iteration samples a different network architecture while sharing weights across all.
- This technique reduces complex co-adaptations, forcing neurons to learn robust features.
- At test time, all neurons are used but their outputs are multiplied by 0.5.
- We apply dropout in the first two fully-connected layers to combat overfitting.



The graph illustrates the training error rate over 40 epochs for two different models or configurations. The solid line represents a model that converges much faster than the model represented by the dashed line. Both models start with a high error rate of 0.75 and decrease as training progresses.

Epochs	Training error rate (Solid Line)	Training error rate (Dashed Line)
0	0.75	0.75
1	0.55	0.65
2	0.45	0.55
3	0.38	0.50
4	0.32	0.45
5	0.28	0.42
6	0.25	0.40
10	-	0.35
15	-	0.32
20	-	0.30
25	-	0.28
30	-	0.26
35	-	0.25

Figure 1 displays a 4x4 grid of 16 image categories. The categories are: mile, container ship, motor scooter, kangaroo, pig, mushroom, cherry, Madagascar cat, giraffe, elephant, lion, tiger, whale, shark, and pumpkin. Each category is represented by a small image and a label.

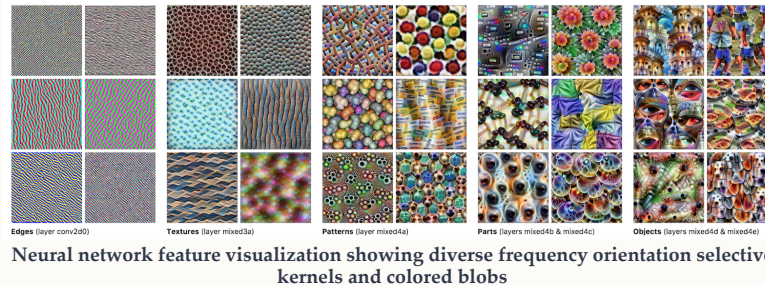
4.1 Performance Comparison with State-of-the-Art

Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
<i>SIFT + FVs [7]</i>	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

Table 1: Comparison of results on ILSVRC-2010 test set. In italics are best results achieved by others.

- Sparse coding models previously achieved 47.1% top-1 and 28.2% top-5 error rates.
- SIFT + FVs approaches reached 45.7% top-1 and 25.7% top-5 error rates.
- Our CNN achieves a new state-of-the-art with 37.5% top-1 and 17.0% top-5 errors.
- This represents a significant improvement over the best previously published results on ILSVRC-2010.
- The results demonstrate the superior performance of deep convolutional networks on large-scale datasets.

4.2 Qualitative Evaluation of Learned Features



- Network learned diverse frequency- and orientation-selective kernels plus colored blobs.
- Two GPUs exhibited specialization: GPU 1 kernels are color-agnostic, GPU 2 kernels are color-specific.
- Even off-center objects, like the mite in the top-left, can be successfully recognized.
- Top-5 labels appear reasonable, such as considering only other cat types plausible for a leopard.
- Feature similarity is measured by Euclidean distance in the 4096-dimensional hidden layer.
- Retrieved training images vary in pose and appearance, yet share high-level semantic similarity.

Thank you for listening!

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